E-COMMERCE PLATFORM USING FULL STACK WEB DEVELOPMENT

INDUSTRIAL INTERNSHIP REPORT

Submitted by

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(An Autonomous Institution | Accredited with 'A++' Grade by NAAC)

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BONAFIDE CERTIFICATE

Certified that this internship report entitled **"E-COMMERCE PLATFORM USING FULL STACK WEB DEVELOPMENT"** is the bonafide work of **"MITHUN P (Reg.No: 921723102087)"**, who carried out the internship work under my supervision.

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Internship Work Presentation held on _____

INTERNAL EXAMINER **EXTERNAL EXAMINER**

ACKNOWLEDGEMENT

I am very thankful to **CODTECH IT SOLUTIONS PRIVATE LIMITED** company for having giving me the opportunity to undertake my internship training at their prestigious company. It was very good learning experience for me to have worked at this site as project involved many unique program development practices and challenges.

I would like to convey my heartiest thanks to **Mr. Neela Santhosh Kumar,** Human Resources & Academic Head of CODTECH IT SOLUTIONS PRIVATE LIMITED, who hearty provided for my internship. The Head who guided and encouraged me all through the internship training and imparted in depth knowledge of the training.

I would like to thank all the department heads of CODTECH IT SOLUTIONS PRIVATE LIMITED for giving their precious time and valuable guidance during my internship programming.

Last but not the least, I would like to thank my Head of the Department, **Dr.B.Lalitha** and I also thank my Coordinator **Mrs.K.Priyadharsini** for allowing me to attend such a training.

ABSTRACT

Sri Murugan E-Commerce is a web-based retail stationery shopping platform developed to simplify the daily shopping and ordering operations of book centres and stationery stores. The system provides a centralized platform for showcasing creative supplies, categorizing educational inventory, managing real-time customer carts, and recording order details. It includes a responsive digital catalog module that allows customers to quickly browse, search, and filter stationery collections across desktop and mobile devices.

The system automatically calculates cart sub-totals, evaluates delivery criteria for free shipping, updates item quantities dynamically, and collects delivery details through an integrated Cash on Delivery (COD) checkout interface. It also provides an organized user interface for monitoring products, viewing category-wise arrangements, and handling cart states securely on the client side.

The application is developed using modern full-stack web technologies, consisting of React.js, Next.js, Node.js, and CSS styling frameworks, along with supporting tools for responsive layout design, state management, and client-side data handling. The system aims to minimize manual counter purchasing queues, enhance product discoverability, save customer time, and provide small retailers with an efficient, scalable, and user-friendly digital storefront for store operations.

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CHAPTER 1
INTRODUCTION

In today's digital environment, technology plays an important role in improving the efficiency of business operations. Small stationery and retail book shops often manage their products, customer orders, stock catalog, and sales records manually through physical visits, which can lead to counter queues, limited reach beyond local boundaries, and time-consuming manual purchasing processes. A web-based e-commerce platform can help overcome these challenges by providing a 24/7 accessible storefront for browsing and purchasing stationery products online.

Sri Murugan E-Commerce is a web-based retail stationery shopping platform developed using modern full-stack web technologies including React.js, Next.js, and Node.js. The system is designed especially for retail book and stationery centers to showcase creative tools, simplify item discovery, streamline cart handling, and collect customer orders digitally.

The application provides features such as an interactive hero showcase ("Elevate Your Creative Daily"), an organized 16-category stationery grid, global product search, a reactive client-side shopping cart, dynamic order total calculation with free shipping rules, and a cash-on-delivery (COD) checkout system. Customers can explore products across stationery categories such as Pens, Notebooks, Planners, Art Kits, and Geometry Sets, add desired quantities to their selection, and submit shipping orders seamlessly.

The platform includes real-time quantity steppers (+ / -), automatic subtotal recalculation, and instant item removal controls. The checkout module captures customer delivery credentials including Name, Email, Phone, and

Address, paired with an order summary card displaying line-item counts and grand totals. Thus, Sri Murugan E-Commerce provides a modern, fast, and user-friendly digital solution for expanding local retail stationery sales while minimizing manual store overhead.

CHAPTER 2
COMPANY PROFILE

2.1 Description about the company

Codtech IT Solutions Private Limited is a technology-based software company and corporate training organization (ISO 9001:2015 Certified) that delivers modern enterprise IT solutions, web applications, and professional technical internships. The company focuses on developing practical technical skills and providing hands-on software development exposure to students through industry-oriented projects.

As part of the internship program, **Full Stack Web Development** was offered to provide practical knowledge of modern web application development. The internship provided exposure to different stages of the software development lifecycle, including frontend user interface engineering, component-driven design, backend integration, state management, and modern development workflows.

The internship was completed during the period **21.01.2026** to **18.02.2026 (Intern ID: CTIS3804)**. Through this internship, practical knowledge of full-stack development was gained by working on a real-world web application project. The experience helped in understanding how frontend interfaces, layout components, state controllers, and checkout forms are integrated to build a complete e-commerce solution.

The internship at Codtech IT Solutions Private Limited provided an opportunity to apply academic knowledge in a practical environment and improve programming, user experience design, debugging, and software implementation skills.

2.2 WORKFORCE

Codtech IT Solutions Private Limited has a team of experienced software developers, technical trainers, and project mentors who support software delivery and provide industry-oriented guidance to trainees. The workforce consists of professional members from different engineering domains who guide interns in understanding modern development workflows and applying technical concepts to real-world applications.

During the internship, the technical mentors provided continuous guidance in Full Stack Web Development, including frontend structure design, component modularity, state reactivity, responsive layout configuration, and project debugging. The interaction with the technical team helped in improving programming skills, understanding agile web practices, and gaining exposure to industry software standards.

The supportive work environment and technical direction from the workforce helped in successfully developing the Sri Murugan E-Commerce Platform as part of the internship.

CHAPTER 3

INTERNSHIP ACTIVITIES

3.1 OBJECTIVE

The main objective of the internship at Codtech IT Solutions Private Limited was to gain practical knowledge and hands-on experience in Full Stack Web Development by applying software development concepts in a real-world project environment.

The specific objectives of the internship were:

- To understand the fundamentals of full-stack web application development.
- To gain practical knowledge of modern frontend frameworks using React.js and Next.js.
- To understand component-based architecture and responsive layout structuring.
- To learn state management techniques for dynamic shopping carts and line-item calculations.
- To improve programming, debugging, and UI design skills.
- To understand the different development stages involved in deploying a digital storefront.
- To develop a practical project, Sri Murugan E-Commerce Platform, using modern web technologies.
- To gain experience in designing a user-friendly and responsive stationery retail catalog.
- To understand how product filtering, shopping carts, and checkout workflows operate together.

• To improve overall technical capabilities and gain exposure to professional software engineering practices.

This internship was a critical step in my academic and career development, allowing me to apply theoretical concepts to real-world software tasks, improve my technical expertise, and build a strong foundation for future growth in web technologies.

3.2 Summary Report

The internship at Codtech IT Solutions Private Limited provided me with an enriching experience in full-stack web development, particularly in the areas of React.js and Next.js for frontend development, component architecture, state management, and responsive UI engineering. The primary project I worked on was Sri Murugan E-Commerce Platform, which was developed to modernize and streamline the retail stationery shopping experience for educational and creative consumers. The project focused on developing a clean user interface, 16-category product organization, interactive cart controls, dynamic total calculation, and a seamless COD checkout flow. Below is a summary of the key components of the project and my contributions:

1. Project Setup and Architecture

Objective: To configure a modern, scalable full-stack development environment for the web application.

Details: I initialized the application using Next.js and React.js running on a local development server (localhost:3000). I configured the folder structure with separate directories for reusable components, page routes, public assets, and styling modules. Global CSS styling rules, fonts, and color palettes were set up to establish a consistent brand aesthetic with warm accents (#F05A22) and deep typography (#1A2238).

2. Landing Page and Brand Showcase

Objective: To build an engaging, modern hero section highlighting store branding and core collections.

Details: I created the primary landing interface featuring the brand identity "Sri Murugan Book Centre & Stationery". The hero banner was designed with the headline "Elevate Your Creative Daily - Curated essentials for the modern learner", along with interactive call-to-action buttons ("Explore Collection") and featured badges ("SMC Originals"). The navigation header was developed with a global search input and an interactive header cart icon displaying active items.

3. Stationery Category Navigation System

Objective: To create an organized, icon-based category navigation system for rapid item discoverability.

Details: I designed and developed a 16-category product grid titled "Tools That Bring Ideas to Life". The grid categorizes retail products into Pens, Notebooks, Planners, Pencils, Erasers, Sharpeners, Sticky Notes, Memo Pads, Calendars, Highlighters, Rulers, To-do Lists, Journals, Bookmarks, Kits, and Geometry Sets. Each card was designed with dedicated iconography, hover responsiveness, and dynamic routing to filter stationery inventory by category.

4. Interactive Shopping Cart Development

Objective: To develop a stateful, interactive cart interface for managing selected items in real time.

Details: I implemented the client-side cart page ("Your Selection"). The interface allows users to view added items, item images, and individual pricing (e.g., Faber-Castell Crayons at ₹55). I engineered dynamic quantity controls (+ / -) allowing instant updates, an item removal button with a trash icon, and

dynamic order summary calculations. Free shipping eligibility rules were programmed to update the delivery fee dynamically to “FREE”.

5. Cash on Delivery (COD) Checkout Workflow

Objective: To implement a streamlined checkout pipeline capturing delivery details and confirming orders.

Details: I developed the dedicated checkout module consisting of two main panels: a Shipping Details input form and an Order Summary panel. The form captures customer identification parameters including Name, Email, Phone, and Address. The summary panel displays ordered items, line quantities, and the calculated total (₹110). A prominent “Place Order (COD)” action button was implemented with validation checking to ensure accurate order placement.

6. Responsive Design and Cross-Device Layouts

Objective: To guarantee consistent performance, responsiveness, and usability across mobile and desktop screens.

Details: Using CSS Flexbox and Grid layouts, I optimized the user interface so that navigation headers, category rows, product cards, and checkout columns adjust fluidly to different screen resolutions. Media breakpoints were integrated to ensure smooth transitions between full desktop monitors and smartphone screens.

Overall, the internship provided valuable practical experience in building a complete full-stack web application. Working on Sri Murugan E-Commerce strengthened my knowledge of React.js, Next.js, component architecture, client-side state handling, form validation, and responsive design, while significantly improving my problem-solving, UI design, and software implementation skills.

CHAPTER 4

DESCRIPTION OF TECHNICAL ASPECTS

4.1 Description about Next.js & React Framework

React.js is an open-source, component-based front-end JavaScript library used for building interactive user interfaces, while Next.js is a production-grade framework built on top of React that provides features such as server-side rendering, route handling, optimized asset compilation, and fast refresh during development.

In the Sri Murugan E-Commerce application, React and Next.js are used to deliver a fast, responsive, and maintainable retail shopping experience. The component-driven structure allows complex interfaces—such as header navigation bars, category cards, cart rows, and shipping forms—to be broken down into modular, reusable building blocks.

4.1.1 Next.js Framework

Next.js provides a streamlined developer environment with built-in page routing, automatic code splitting, and optimized asset handling. The application runs on a local development server on port 3000 (localhost:3000), allowing instantaneous hot-reloading when source files are updated.

In the Sri Murugan project, Next.js page routing manages transitions between the primary store landing page (/), category listing views (/shop/category/[slug]), the interactive cart interface (/cart), and the checkout portal (/checkout). Static assets, such as stationery icons and branding logos, are served efficiently through the public directory, reducing loading latency.

4.1.2 React State Management

State management is a core technical aspect of the Sri Murugan E-Commerce system. React's built-in state hooks (useState and useEffect) are used to manage data changes on the client side without requiring manual browser reloads.

In this project, client-side state is responsible for:

- Maintaining the active list of selected items in the cart.
- Tracking item quantity increments and decrements in real time.
- Synchronizing the shopping cart counter badge in the header across all views (e.g., Shop 2).
- Managing input form values (Name, Email, Phone, Address) during checkout.

When a user adjusts the quantity of an item like Faber-Castell Crayons, the cart state updates immediately, triggering automatic recalculation of line totals and order sums.

4.1.3 Component Architecture

The application is structured into modular, reusable components to ensure clean code organization and maintainability. The key components developed for the system include:

- **Navigation Bar Component:** Houses the store logo, category dropdown menu, global search input field, and reactive cart badge.
- **Hero Banner Component:** Displays promotional headlines, collection highlights, and action buttons.
- **Category Grid Component:** Maps stationery department arrays into responsive interactive cards.

• **Cart Selection Component:** Displays line-item cards with image thumbnails, price tags, quantity steppers, and remove triggers.

• **Checkout Form Component:** Organizes customer address fields and presents the final order summary before purchase confirmation.

4.1.4 Client-Side Cart & CRUD Operations

CRUD operations represent the four fundamental data operations: Create, Read, Update, and Delete. In the Sri Murugan E-Commerce application, these operations are implemented on the client side to manage shopping cart contents:

Create: The Create operation occurs when a customer selects a stationery product from the category or product showcase. The item details, including name, unit price, thumbnail image, and initial quantity, are added to the active cart state array.

Read: The Read operation retrieves and renders the cart data on both the dedicated cart screen (“Your Selection”) and the checkout summary panel. It reads line items, calculates the subtotal, checks free-shipping qualification, and displays the grand total.

Update: The Update operation modifies existing item quantities. Using the plus (+) and minus (-) stepper buttons on the cart interface, users can increment or decrement item units. The system updates the item quantity and immediately recalculates the corresponding total amount.

Delete: The Delete operation removes an item from the cart state entirely. When the trash icon button next to an item is clicked, that item is filtered out from the cart array, and the header badge count and overall total are updated accordingly.

CHAPTER 5

SCREENSHOTS

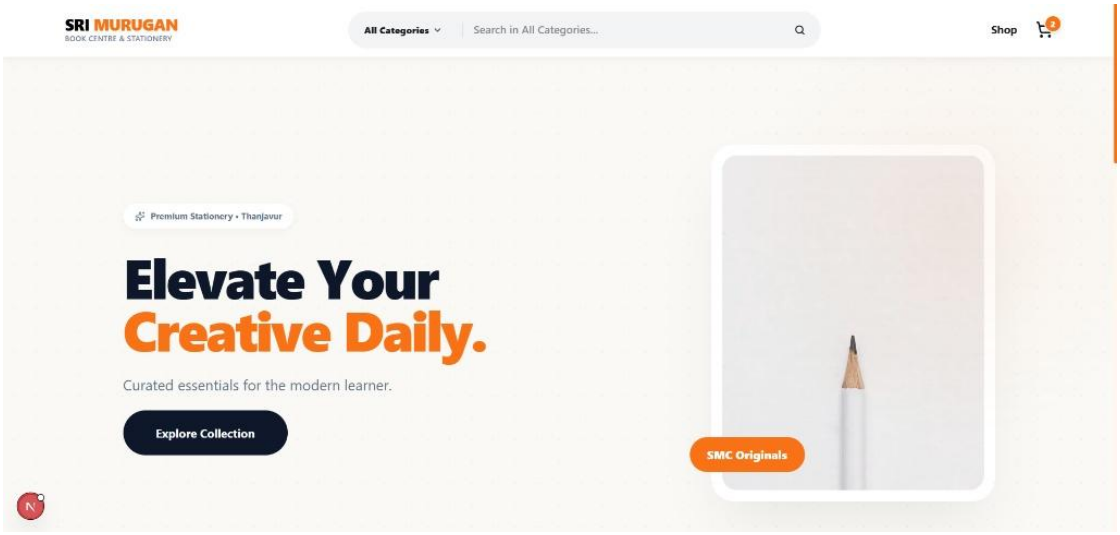


Figure 5.1: Sri Murugan Book Centre & Stationery - Landing Interface with Brand Hero Banner, Search Bar, and Cart Badge

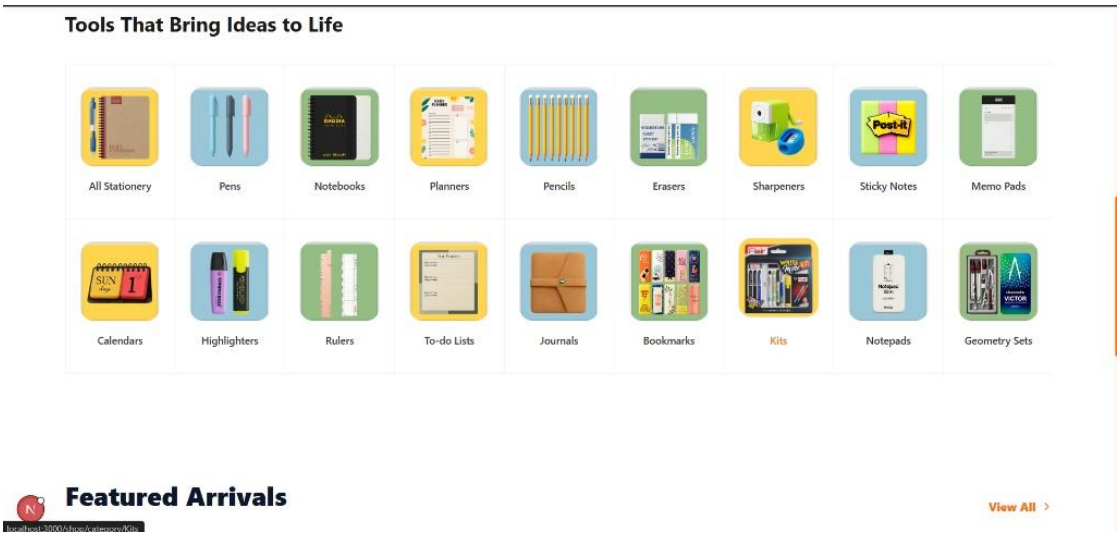


Figure 5.2: Stationery Category Grid displaying 16 categorized retail stationery departments ("Tools That Bring Ideas to Life")

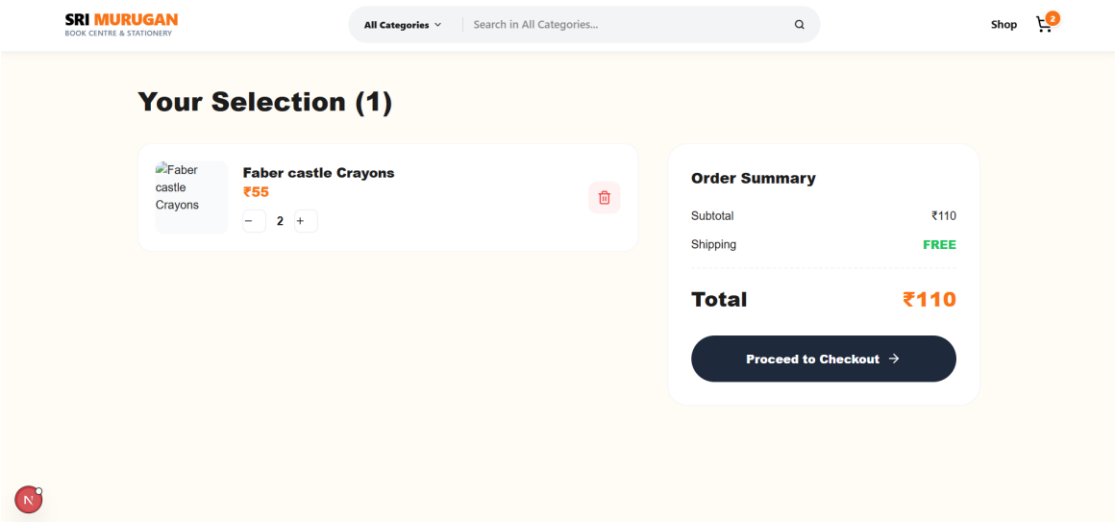


Figure 5.3: Cart Selection Interface ("Your Selection") with item quantity modifiers, price calculations, and free shipping

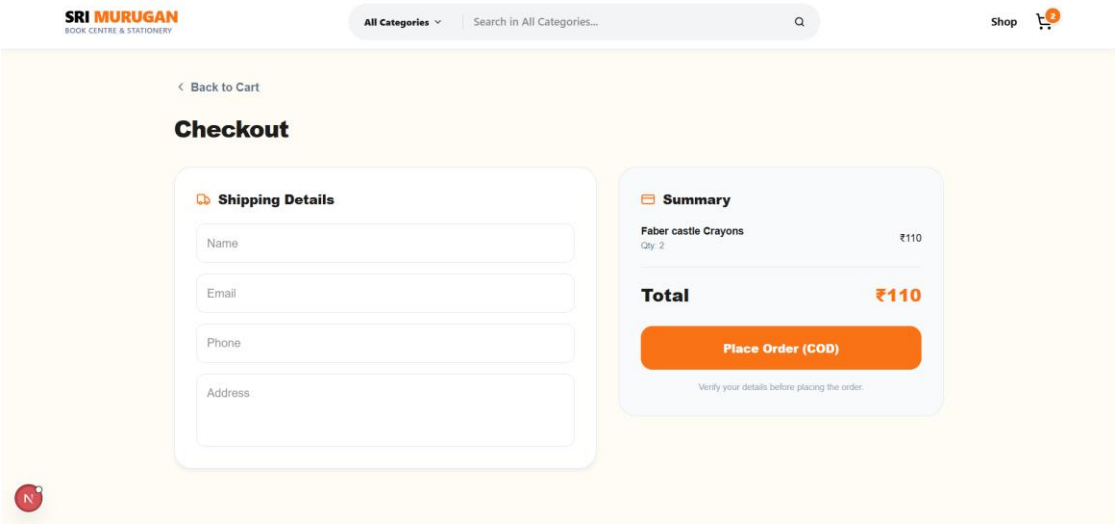


Figure 5.4: Checkout Interface displaying Shipping Details Form, Order Summary Card, and Place Order (COD) Button

CHAPTER 6

CONCLUSION

The internship at Codtech IT Solutions Private Limited provided valuable practical experience in Full Stack Web Development and helped in understanding the complete process of building a modern web application. During the internship, the Sri Murugan E-Commerce Platform was successfully developed using modern web technologies including React.js, Next.js, and Node.js.

The project successfully integrates essential features required for a digital retail stationery storefront. The system provides an interactive brand landing page, a 16-category product navigation grid, dynamic product search, client-side shopping cart management with real-time total updates, free shipping rules, and a streamlined Cash on Delivery (COD) checkout interface.

The application leverages component-driven frontend architecture, modular CSS styling, dynamic client-side state handling, and structured checkout form processing. The development process improved my understanding of responsive design, state reactivity, CRUD operations in client state, debugging, and local server management.

Overall, the internship was a valuable learning experience that strengthened both technical and professional engineering skills. The successful completion of the Sri Murugan E-Commerce Platform demonstrated the practical application of full-stack web development principles to solve real-world retail challenges, providing a strong foundation for future software engineering endeavors.

INTERNSHIP CERTIFICATE



CODTECH IT SOLUTIONS PRIVATE LIMITED

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CERTIFICATE

OF INTERNSHIP EXPERIENCE

To whom so ever it may concern

Intern Name: Mithun P
Intern ID: CTIS3804
Program Type: Internship
Duration: 4 Weeks
Domain: Full Stack Web Development
Organization: Codtech IT Solutions Private Limited
Internship Period: 21 January 2026 - 18 February 2026
Status: Successfully Completed
Remarks: Demonstrated strong dedication, technical skills, and a positive learning attitude throughout the internship.

We appreciate his/her active participation, consistent efforts, and impressive contribution to the overall success of the internship.We wish him/her all the best in future endeavors.

We are confident that his/her dedication and skills will lead to great achievements ahead.

Best Wishes,
NEELA SANTHOSH KUMAR
Human Resources & Academic Head
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